

Models config						
Model	Strain features	Feature selection strategy	Evaluation scheme	Accuracy	ROC-AUC	F1-weighted
HALO-CV2	Yes	Nested (within folds)	CV2 (pair + strain held-out)	0.50	0.52	0.48
HALO-CV1	Yes	Nested (within folds)	CV1 (pair-held-out)	0.68	0.74	0.68
HALO	No	Nested (within folds)	CV1 (pair-held-out)	0.68	0.74	0.68
HALO-PreCV	No	Global (pre-CV)	CV1 (pair-held-out)	0.75	0.82	0.75
HALO-BASE	No	Training-set only	Standard stratified split	0.76	0.84	0.76